

17/00

1) Given:file size = 150×10^6 bytes

segment size = 1500 bytes

data link overhead = 40 bytes

i) Number of segments

$$\frac{150\,000\,000}{1500} = 100\,000 \text{ segments}$$

ii) Number of frames

Each frame segment is encapsulated into one frame

$$\text{Frames} = 100\,000$$

iii) Total Data link overhead

$$100\,000 \times 40 = 4\,000\,000 \text{ bytes} = 4 \text{ MB}$$

iv) Total Data Transmitted

$$150 + 4 = 154 \text{ MB}$$

convert to bits

$$154 \times 10^6 \times 8 = 1\,232\,000\,000 \text{ bits}$$

Transmission time

$$\frac{1\,232\,000\,000}{50 \times 10^6} = 24.64 \text{ s}$$

Reliability:

- Transport layer: acknowledgements, error recovery, flow control.
- Data link layer: MAC addressing, retransmission of damaged frames.

2) Given:

window size = 6 frames

total frames = 48

i) Number of windows

$$\frac{48}{6} = 8$$

ii) Retransmissions

one frame lost in every window

$$8 \times 1 = 8$$

Flow control

Flow control prevents sender from overwhelming the receiver, avoids buffer overflow, improves throughput and network efficiency.

3) Given:

IP packet = 12000 bytes

MTU = 1500 bytes

IP header = 20 bytes

Payload per Fragment

$$1500 - 20 = 1480 \text{ bytes}$$

Number of Fragments

$$\left\lceil \frac{12000}{1480} \right\rceil = \lceil 8.108 \rceil = 9$$

Header overhead

$$9 \times 20 = 180 \text{ bytes}$$

Network layer role

uses fragmentation, identification field, routing and reassembly at destination for successful packet delivery.

Given:

windows size = 6 frames

total frames = 48

i) Number of windows

$$\frac{48}{6} = 8$$

ii) Retransmissions

one frame lost in every window

$$8 \times 1 = 8$$

Flow control

Matches sender speed with receiver capability, preventing congestion and packet loss

5) Given:

file = 600 MB

checkpoint every 60 MB

failure after 390 MB

checkpoints before failure

$$\frac{390}{60} = 6.5$$

Completed Checkpoints

60, 120, 180, 240, 300, 360

Retransmission

last checkpoint = 360

failure = 390

$$390 - 360 = 30 \text{ MB}$$

Importance:

Synchronization allows communication to resume from the checkpoint instead of restarting the entire transfer.

6) Given:

original file = 900 MB

Compression = 40%.

Encryption overhead = 5%.

Compressed Size

$$900 \times 0.6 = 540 \text{ MB}$$

Encrypted Size

$$540 \times 1.05 = 567 \text{ MB}$$

Overall Reduction

$$\frac{900 - 567}{900} \times 100 = 37\%$$

Presentation layer role:

Provides compression, encryption, decryption, data format conversion; improve efficiency and security.

7) Given:

File = 3 GB = 3000 MB

Request size = 3 MB

Throughput = 120 Mbps

Requests

$$\frac{3000}{3} = 1000$$

Transmission Time

in bits

$$3000 \times 10^6 \times 8 = 24 \times 10^9 \text{ bits}$$

$$\frac{24 \times 10^9}{120 \times 10^6} = 200 \text{ s}$$

Application layer role:

Provides services such as FTP, authentication, file transfer, user interface and application-level error handling.

Given:

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data link = 2 ms

Physical = 1 ms

Propagation = 18 ms

Transmission = 12 ms

Total Delay:

$$4 + 3 + 5 + 2 + 1 + 18 + 12 = 45 \text{ ms}$$

Layer Contributions

- Application - user services
- Transport - Reliable delivery
- Network - Routing
- Data link - Framing and error detection
- Physical: Bit transmission

9) Given:

useful data = 80 MB = 80000000 bytes

frame size = 1600 bytes

overhead = 80 bytes

useful payload/frame:

$$1600 - 80 = 1520 \text{ bytes}$$

Total Frames:

$$\left[\frac{80000000}{1520} \right] = 52632 \text{ frames}$$

Total Overhead

$$52632 \times 80 = 4210560 \text{ bytes}$$

Transmission Efficiency

$$\frac{1520}{1600} \times 100 = 95\%$$

Effect of Overload

Higher protocol overhead reduces available bandwidth, lowers efficiency and increases transmission time.

10) Given:

original file = 240 MB

compression = 30%

segment size = 1400 bytes

frame overhead = 60 bytes

i) Compressed size

$$240 \times 0.70 = 168 \text{ MB}$$

$$= 168000000 \text{ bytes}$$

ii) Number of segments

$$\frac{168000000}{1400} = 120000$$

iii) Total overhead

$$120000 \times 60 = 7200000 \text{ bytes} = 7.2 \text{ MB}$$

iv) final Data sent

$$168 + 7.2 = 175.2 \text{ MB}$$

working of Multiple OSI Layer:

- presentation layer - compression and encryption
- Transport layer - segmentation, flow control and reliability
- Network layer - logical addressing and routing
- Data link layer - Framing and error detection
- Physical layer - Transmits the bits over the communication medium.

Qubits
20MB